DIFFERENTIAL -TRAC-LOK (Continued)

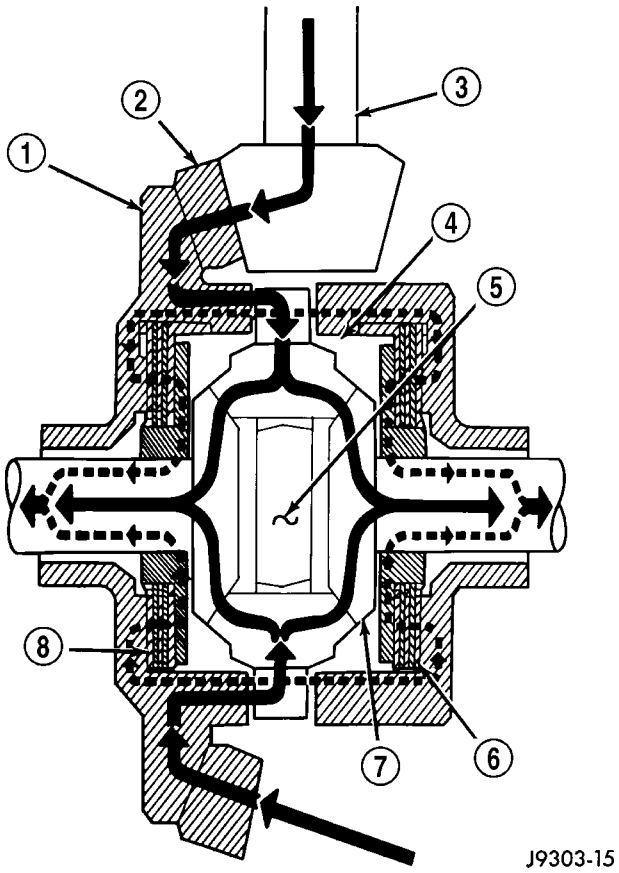

Fig. 48 TRAC-LOK LIMITED SLIP DIFFERENTIAL

1 - CASE
2 - RING GEAR
3 - DRIVE PINION
4 - PINION GEAR
5 - MATE SHAFT
6 - CLUTCH PACK
7 - SIDE GEAR
8 - CLUTCH PACK

DIAGNOSIS AND TESTING

The most common problem is a chatter noise when turning corners. Before removing the unit for repair, drain, flush and refill the axle with the specified lubricant. A container of Mopar Trac-lok® Lubricant (friction modifier) should be added after repair service or during a lubricant change.

After changing the lubricant, drive the vehicle and make 10 to 12 slow, figure-eight turns. This maneuver will pump lubricant through the clutches. This will correct the condition in most instances. If the chatter persists, clutch damage could have occurred.

DIFFERENTIAL TEST

The differential can be tested without removing the differential case by measuring rotating torque. Make sure brakes are not dragging during this measurement.

(1) Place blocks in front and rear of both front wheels.

(2) Raise one rear wheel until it is completely off the ground.

(3) Engine off, transmission in neutral, and parking brake off.

(4) Remove wheel and bolt Special Tool 6790 or equivalent tool to studs.

(5) Use torque wrench on special tool to rotate wheel and read rotating torque (Fig. 49).

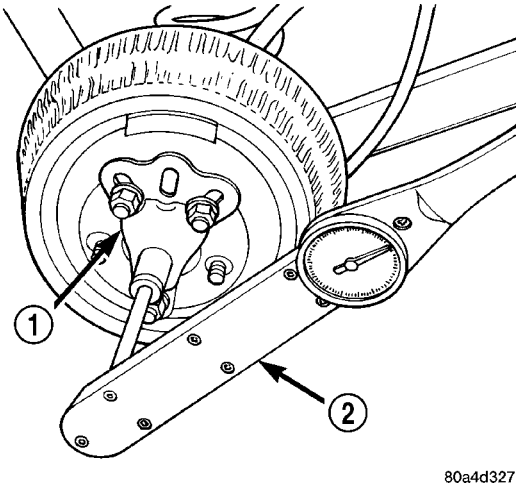

Fig. 49 ROTATING TORQUE TEST

1 - SPECIAL TOOL WITH BOLT IN CENTER HOLE
2 - TORQUE WRENCH

(6) If rotating torque is less than 41 N·m (56 ft. lbs.) or more than 271 N·m (200 ft. lbs.) on either wheel the unit should be serviced.

DISASSEMBLY

(1) Clamp side gear Fixture 6965 in a vise and set differential case on the fixture (Fig. 50).

NOTE: The Trac-lok® differential can be serviced with the ring gear installed unless the gear hangs over the pinion shaft, then it must be removed.

(2) Remove pinion gear mate shaft roll pin with a hammer and punch.

(3) Remove pinion gear mate shaft with a drift and hammer.

(4) Lubricate and install disc without threaded hole from Trac-lok® Tool C-4487 into lower side gear (Fig. 51).

(5) Lubricate and install disc with threaded hole from Trac-lok® Tool C-4487 into upper side gear. Thread forcing screw from tool into threaded disc until it becomes centered in lower disc.

(6) Position a small screw driver in slot of the threaded disc (Fig. 52) to prevent adapter from turning.

(7) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress Belleville springs in clutch packs (Fig. 53).